

Fig. 2: Actual-size PCB layout for the battery charger and discharger with USB power supply

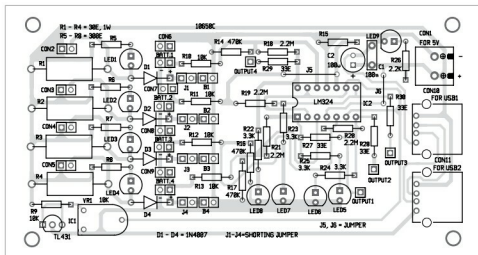


Fig. 3: Component layout for the PCB

resistors R1 through R4 are around 30-ohm, but you can choose any appropriate value. You can also remove R1 through R4, if desired. Resistors R5 through R8 protect LED1 through LED4. The light emitted by these LEDs is proportional to the charging current.

Construction and testing

An actual-size, single-side PCB layout for the battery charger/discharger with USB power supply is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on a PCB, connect CON1 to 5V power supply. Connect bulbs B1 through B4 on the front panel to indicate the discharging status. **EFY**

Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia, Bulgaria, and expert-lecturer in OFPPT (Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria.



Read more stories on solar at
www.electronicb2b.com

TOP
SOLAR STORIES

- Solar lantern
- Inverters for solar installation
- India's regional solar market
- Solar energy for your home

ELECTRONICS
B2B
INDUSTRY IS AT A

CLICK OF A BUTTON

Log on to www.electronicb2b.com and be in touch with the Electronics B2B Fraternity 24x7